

PLS-SEM: Prediction-oriented solutions for HRD researchers

Amanda E. Legate¹  | Joe F. Hair Jr² |
Janice Lambert Chretien³  | Jeffrey J. Risher⁴

¹Department of Human Resource Development, University of Texas at Tyler, Tyler, Texas, USA

²Department of Marketing & Quantitative Methods, University of South Alabama, Mobile, Alabama, USA

³Soules College of Business, University of Texas at Tyler, Tyler, Texas, United States

⁴Department of Management and Marketing, Southeastern Oklahoma State University, Durant, Oklahoma, USA

Correspondence

Amanda E. Legate, Department of Human Resource Development, University of Texas at Tyler, Tyler, TX, USA.

Email: alegate@patriots.uttyler.edu

Abstract

Structural equation modeling, often referred to as SEM, is a well-established, covariance-based multivariate method used in Human Resource Development (HRD) quantitative research. In some research contexts, however, the rigorous assumptions associated with covariance-based SEM (CB-SEM) limit applications of the method. An emergent complementary SEM approach, partial least squares structural equation modeling (PLS-SEM), is a variance-based SEM method that provides valid solutions and overcomes several limitations associated with CB-SEM. Despite PLS-SEM's increasing popularity in many social sciences disciplines, the method has yet to gain traction in the field of HRD. An accessible overview of the method, including potential advantages for HRD research and extant methodological advancements, is provided in this article with the goal of encouraging productive dialogue in the field of HRD surrounding the PLS-SEM approach. We present an emergent analytical tool for quantitative HRD research, offer practical guidelines for researchers to consider when selecting a SEM method, and clarify assessment stages and up-to-date evaluation criteria through an illustrative example.

KEYWORDS

confirmatory composite analysis, human resource development, partial least squares, PLS-SEM, variance-based SEM

1 | INTRODUCTION

In the social sciences, structural equation modeling (SEM) is a well-established, multivariate statistical approach for examining relationships among unobservable (i.e., latent) variables (Willaby et al., 2015). Since its introduction in the 1970s, SEM has quickly become a method of choice for many scholars when analyzing complex models (Jöreskog & Wold, 1982; Kline, 2016). Contemporary literature describes two approaches to SEM, which offer fundamentally unique solutions for modeling these relationships. Often referred to simply as SEM (Kline, 2016; Schumacker & Lomax, 2016), the more commonly known covariance-based SEM (CB-SEM) has a longer history of widespread research application in the social sciences (Hair, Risher, et al., 2019; Ringle et al., 2020). A complementary SEM approach, partial least squares structural equation modeling (PLS-SEM; Jöreskog & Wold, 1982), has gained increasing popularity in organizational research applications over the past decade (Hair et al., 2022).

SEM refers to a family of related statistical methods and procedures that combine elements of regression and correlation analyses in a way that enables researchers to simultaneously analyze relationships among variables (Huck, 2012; Kline, 2016). Explanatory SEM modeling (as with CB-SEM) focuses on estimating a theoretical model that fits the sample data as well as testing hypothesized relationships among constructs (Schumacker & Lomax, 2016). CB-SEM solutions are obtained using common variance and the maximum-likelihood (ML) estimation approach, with the goal of minimizing the differences between the observed and estimated covariance matrices (Hair & Sarstedt, 2021). In PLS-SEM, solutions are not driven by comparing theorized and data-implied correlations (i.e., covariances), but rather PLS-SEM derives solutions based on total (i.e., common and unique) variance with the objective of jointly minimizing the residuals in the measurement models and the structural model (Lohmöller, 1989; Manley et al., 2020; Wold, 1982). In the following pages, we identify differences and similarities between PLS-SEM and other multivariate techniques, namely, CB-SEM. We preface such comparisons with an acknowledgment that comparative reference points are intended to present PLS-SEM as a complementary, rather than a competing, approach (Jöreskog & Wold, 1982). Indeed, we agree with Rigdon et al. (2017) that comparing methods in terms of which is the better approach bypasses the point of productive methodological dialogue. As such, the purpose of this article is to (a) cultivate awareness among HRD researchers about PLS-SEM; (b) provide an accessible overview of the PLS-SEM methodology for a broad audience of HRD scholars and practitioners; and (c) demonstrate PLS-SEM assessment stages and evaluation criteria through an illustrative example.

2 | PLS-SEM IN BUSINESS AND ORGANIZATIONAL RESEARCH

The PLS-SEM approach has become increasingly popular in a variety of business and organizational research contexts. Numerous studies have demonstrated its widespread application in fields such as accounting (Lee et al., 2011; Nitzl, 2016), entrepreneurship (Manley et al., 2020), family business (Sarstedt, Ringle, Smith, et al., 2014), higher education (Ghasemy et al., 2020), human resource management (Ringle et al., 2020), international business (Richter et al., 2016), knowledge management (Cepeda-Carrión et al., 2019), management (Hair et al., 2012), management information systems (Ringle et al., 2012), and marketing (Hair et al., 2022), to name a few. In addition, several popular peer-reviewed journals such as *International Marketing Review* (Richter et al., 2016), *Journal of Business Research* (Cepeda-Carrión et al., 2016), *Journal of Marketing Theory & Practice* (Hair et al., 2011), and *Long Range Planning* (Sarstedt, Ringle, & Hair, 2014) have dedicated entire special issues to the PLS-SEM method.

Despite evidence of the method's popularity and an extensive body of emerging methodological advancements, there appears to be a lack of awareness about PLS-SEM in the field of HRD. In preparing for this article, we reviewed quantitative studies published in *Human Resource Development Quarterly* (HRDQ) to examine the extent to which various analytic techniques were employed by HRD researchers over the past decade (2010 to April 2021). Of the 93 HRDQ articles reviewed, SEM was predominantly reported, comprising 37% of all quantitative analyses conducted during this period, followed closely by multiple regression (25%). All SEM articles reviewed applied CB-SEM; to date, no

quantitative study published in *HRDQ* has reported PLS-SEM. Given the prolific application of PLS-SEM in tangentially connected disciplines, this result was surprising. On the other hand, this finding may not be entirely unexpected given that throughout its nascency, the PLS-SEM approach has been the topic of nuanced scholarly debate. Admittedly, providing explanations for a delayed adoption by HRD researchers was not within the scope of the present study. It is possible, however, that HRD manuscripts reporting PLS-SEM have not been accepted for publication due to a lack of satisfactory methodological justifications and/or because relatively little is known about the method within the field.

In keeping with our stated purpose, this manuscript is organized into four main sections. In the first section, we present an overview of the PLS-SEM approach, highlighting unique features of the method. Second, we address contemporary dialogue surrounding methodological justifications and comparison-based criticisms. Third, we describe relevant distinguishing characteristics of PLS-SEM's measurement and structural philosophy. In the fourth section, we demonstrate the application of PLS-SEM assessment stages and evaluation criteria through an illustrative model. We conclude with a brief discussion.

3 | THE PLS-SEM APPROACH

In HRD empirical research, SEM is commonly used to denote the covariance-based approach or CB-SEM. Though less common in *HRDQ* studies, in other disciplines, PLS-SEM is often considered a viable analytical approach (for an exception, see Akdere & Egan, 2020). PLS-SEM offers several relatively unique features when considering its application including extended flexibility related to data characteristics, suitability for exploratory theory-building research, and enhanced prediction metrics (Hair & Sarstedt, 2021; Shiau et al., 2019; Streukens & Leroi-Werelds, 2016). As is often fundamental to promoting methodological advancement (see Zhao et al., 2010), PLS-SEM has received a fair share of critical reflection and scholarly discourse. Discussions have focused mostly on the method's differences compared to CB-SEM, including the lack of global fit measures and treatment of measurement error (McIntosh et al., 2014; Rigdon, 2016; Rönkkö et al., 2016; Rönkkö & Evermann, 2013; Sarstedt, Ringle, & Hair, 2014). Furthermore, methodologists have commented on situations in which characteristics of the PLS-SEM method, primarily related to small sample sizes and robust solutions with nonnormal data, have been overstated and abused (Goodhue et al., 2012; Hair et al., 2017; Marcoulides et al., 2012; Ringle et al., 2020). At the same time, this emergent method has experienced rapid development across numerous fields in recent years (Hair, Risher, et al., 2019; Manley et al., 2020). With enhanced understanding, the method's features may be beneficial to HRD researchers.

3.1 | Extended flexibility

A particularly valuable advantage of PLS-SEM is its ability to utilize not only metric data, but also nonmetric (i.e., nominal and ordinal), and even binary data when coded as dummy variables (Hair et al., 2022). In addition, it is the preferred approach when a theoretical model includes formatively measured constructs developed based on observational and secondary data (Hair, Black, et al., 2019; Sarstedt et al., 2016). In empirical HRD research, scholars and practitioners call for inclusion of a variety of objective measures, not only to offset correlational disturbances attributed to common methods (Podsakoff et al., 2003), but to refine and/or extend existing empirical contributions. In addition to self- and other-rated appraisals, HRD researchers also call for the inclusion of more objective measures of performance outcomes (Carter & Youssef-Morgan, 2019), physical outcomes (Osam et al., 2020), and behavioral outcomes (Zigarmi et al., 2015). While employee self-reports are often ideal sources for subjective psychological appraisals, this may not always be the case for key constructs of interest in HRD research. Whereas the type of scale is particularly influential in CB-SEM analyses and requires special considerations (Kline, 2016), PLS-SEM more easily accommodates variables measured using a variety of data types and sources (Hair et al., 2022).

PLS-SEM also offers HRD researchers enhanced flexibility in research design without compromising model specification. First, PLS-SEM is well suited for theoretically exploring model extensions and considered an appropriate methodological choice when identifying principal drivers of target outcome variables is a primary research focus (Hair et al., 2022; Shmueli et al., 2016). In addition, for some research contexts, a large number of estimated parameters can limit analytic options for researchers conceptualizing and testing new models in HRD (see Takeuchi et al., 2020), but this is not a limitation of PLS-SEM. Indeed, in some cases, the PLS-SEM method may offer HRD researchers an appropriate methodological solution when complexity alone limits the application of CB-SEM (Hair et al., 2022).

Regression-based techniques such as conditional process analysis (i.e., PROCESS; Hayes & Rockwood, 2020) execute mediation analyses but are subject to many of the same limitations as ordinary least squares (OLS) regression (Sarstedt et al., 2020). As noted by HRD researchers, a limiting feature of regression is the inability to simultaneously test multiple outcomes (Carter & Youssef-Morgan, 2019; Goldberg et al., 2019). PLS-SEM can assist HRD researchers in adopting a methodology appropriate for exploring theoretically established model extensions and complex integrative models that involve the estimation of many parameters (Ringle et al., 2020). In short, the ability of PLS-SEM to assess multiple outcome variables simultaneously enables researchers to estimate direct, indirect (mediation), and interaction (moderation) effects while at the same time removing measurement error, thus eliminating the need for ad hoc regression analyses (Hair et al., 2022; Sarstedt et al., 2019).

3.2 | In-sample and out-of-sample prediction

In most social sciences disciplines, researchers generally have not focused on prediction designed to infer to the population (Shmueli, 2010). Instead, they have reported coefficients of determination (R^2) and similar prediction metrics that provide evidence of the explanatory characteristics as well as in-sample prediction. That is, researchers typically evaluate whether model coefficients are significant and in the hypothesized direction and report the R^2 value of their model (Hair & Sarstedt, 2021; Hair, Sarstedt, et al., 2019). Computation of the R^2 draws on the estimates produced by the entire dataset to predict the dependent variables' data that has already been used to obtain an optimal statistical solution. This focus on in-sample prediction is useful, but limiting, since it essentially explains the relationships between the modeled variables but does not assess the extent to which a theoretical model can infer to the population (out-of-sample prediction), which lies at the heart of most social sciences research (Popper, 1962).

Recent methodological developments have extended the predictive capabilities of PLS-SEM beyond the traditional explanatory and in-sample model prediction metrics (R^2 and f^2). Specifically, PLS-SEM developments include expanded out-of-sample prediction metrics to assess theoretical models (Hair & Sarstedt, 2021). As demonstrated later in this article, to obtain out-of-sample prediction metrics researchers must first use an initial training sample to estimate model parameters and second, apply those initial parameters to predict values of the dependent variables in a second hold-out sample (Shmueli et al., 2019). The process of using one sample to develop model parameters and predicting the dependent variable in a second sample is referred to as out-of-sample prediction. To implement these predictive assessment techniques, HRD researchers can apply methodological features readily available with SmartPLS software (Ringle et al., 2015; Shmueli et al., 2019).

4 | CONTEMPORARY DIALOGUE

Although justifications for the application of PLS-SEM have gradually become more grounded in well-considered reasons, early justifications for applying PLS-SEM were predominantly related to sample size or lack of normally distributed data (Khan et al., 2019; Rigdon, 2016; Ringle et al., 2020). More recent assessments of PLS-SEM view the method as a robust approach for estimating complex models when compared to the more rigorous statistical assumptions associated with CB-SEM (Shiau et al., 2019; Wold, 1982). The methodological straightforwardness of

the nonparametric PLS-SEM approach, however, should not be equated with a lack of rigor. For extensive comparisons of CB-SEM and PLS-SEM, also see Jöreskog and Wold (1982), Lohmöller (1989), Sarstedt et al. (2016), and Rigdon et al. (2017). As sample size and distributional assumptions are closely related considerations (Kline, 2016), the subsequent section provides an overview of PLS-SEM characteristics, which result in additional flexibility when viewed alongside CB-SEM or regression techniques. This is followed by discussion of critiques surrounding PLS-SEM's inability to imitate CB-SEM. A comprehensive review of multivariate statistical assumptions and multiple linear regression is not feasible in this article; if needed, we recommend the Nimon (2012) mini-review for refresher on key statistical assumptions across the general linear model and the guidebook of variable importance by Nathans et al. (2012) for interpreting multiple regression results.

4.1 | Sample size and distributional considerations

Sample size considerations for PLS-SEM differ from those of CB-SEM, resulting in PLS-SEM's ability to accommodate a greater range of sample sizes. For the family of CB-SEM techniques, larger sample sizes are often required to maintain adequate statistical power, obtain stable parameter estimates, meet identification requirements, and obtain results for more complex models with many indicators (Fornell & Bookstein, 1982; Schumacker & Lomax, 2016). CB-SEM applications require target respondent-to-indicator ratios determined by the total number of indicators in the measurement model, typically following a ratio of 5:1 and in some instances 10:1 (Kline, 2016). While CB-SEM analyzes all the variables in the model simultaneously to obtain solutions, PLS-SEM estimates partial regression models by using an algorithm that iterates between the measurement model and structural model estimations with the goal of minimizing the residuals and optimizing variance explained (Wold, 1982). Plainly stated, because the PLS-SEM method estimates the model's partial regressions instead of estimating the entire model at the same time, smaller sample sizes are required (see Hair et al., 2022, for further explanations of the PLS-SEM algorithm).

There are several approaches to determine the minimum sample size required (Hair et al., 2022). A particularly suitable method is the inverse square root approach proposed by Kock and Hadaya (2018). This approach considers the probability that the ratio of a path coefficient, and its standard error will be greater than the critical value of a test statistic for a given significance level. Therefore, the results for the minimum sample size technically required depend only on a path coefficient and not on the size of the most complex regression in the model. For example, assuming a significance level of 5% and a minimum path coefficient of 0.2, the minimum sample size is 155. While PLS-SEM accommodates a range of sample sizes, it should not be considered a small-sample alternative to CB-SEM. As Rigdon (2016) noted, difficulty in obtaining an adequate sample size is seldom a methodological justification, rather, it is "the nature of the population that justifies the small sample size" (p. 600). For example, in business-to-business research, samples are often constrained by smaller population sizes. Likewise, HRD researchers are often interested in exploring phenomena among more narrowly defined populations, such as human resources professionals (Goldberg et al., 2019), knowledge workers (Jeong, 2021), and university faculty (Hutchins et al., 2018). In such cases, PLS-SEM can often achieve meaningful solutions for both simple and complex models (Hair & Sarstedt, 2021).

When possible larger sample sizes are preferred for any analytic approach to substantiate the ability to infer sample results to a relevant population (Kock & Hadaya, 2018). However, when measuring unobservable phenomena, choice of instrumentation is fundamentally essential to researchers' goals (Fowler, 2014). In such scenarios, PLS-SEM enables researchers to analyze complex models without compromising other key elements of study design. For example, Osam et al. (2020) measured workplace climate with the 21-item Psychological Climate Measure (PCM; Brown & Leigh, 1996). Following the inverse square root approach and assuming a significance level of 1%/5%/10% for a path coefficient between 0.11 and 0.20 results in a minimum sample size of 251/155/113 observations for this set of parameters. Since $n = 259$ respondents were available in this study, PLS-SEM can reliably render the corresponding effects between 0.11 and 0.2 (and higher) significant. In this situation, PLS-SEM can leverage its flexibility to obtain robust results with a smaller sample size.

A distributional characteristic of the nonparametric PLS-SEM method is its ability to obtain robust model estimations with nonnormal and highly skewed data (Hair, Black, et al., 2019). When data do not meet the assumption of normality in CB-SEM, larger sample sizes are often required to obtain solutions (Hair et al., 2017; Kline, 2016). As a methodological justification, this feature provides limited support for choosing PLS-SEM since well-documented remedies exist for CB-SEM when multivariate normality, assumed by ML estimation, is not met (i.e., nonparametric bootstrapping; Kline, 2016). One situation in which distributional flexibility is a comparative benefit for PLS-SEM is when the lack of normality/skewness is combined with a sample size too small for CB-SEM solutions but acceptable for PLS-SEM to obtain a solution (Fornell & Bookstein, 1982).

4.2 | Comparison-based critiques

Critiques surrounding PLS-SEM have identified several methodological differences when compared with CB-SEM, including the lack of global fit indices and alleged failure to remove measurement error (Rönkkö et al., 2016). The claim that PLS-SEM does not imitate CB-SEM is accurate in a general sense. Researchers credited with the development CB-SEM and PLS-SEM asserted that the two methods should not be viewed as competing, but rather, as complementary approaches (Jöreskog & Wold, 1982; Willaby et al., 2015). Herman Wold, recognized for founding modern econometric methods, dedicated 30 years to the development of PLS-SEM's underlying algorithm because he believed that distributional assumptions and limited model complexity imposed by ML estimation were unrealistic for many questions arising in social sciences research (Dijkstra, 2010; Hendry et al., 1994; Wold, 1982). With fundamentally different optimization algorithms and statistical objectives, PLS-SEM does not mimic CB-SEM because it was not developed for that purpose (Jöreskog & Wold, 1982; Lohmöller, 1989). To encourage dialogue above and beyond a competing methods perspective, the following section provides an overview of the measurement and structural philosophy that guides and differentiates the PLS-SEM approach.

5 | PLS-SEM MEASUREMENT AND STRUCTURAL PHILOSOPHY

From a path diagram perspective, PLS-SEM and CB-SEM are similarly comprised of two distinct models (see Figure 1): the measurement model (*outer model* in PLS-SEM), representing the constructs and their associated observable indicators, and the structural model (*inner model* in PLS-SEM), depicting the structural relationships between the constructs in the path model (Shmueli et al., 2016). Unlike multiple regression, both methods provide a powerful set of analytical tools for simultaneous modeling of multiple unobserved (latent) variables and assessment of relationships between multiple independent and dependent variables. Both approaches begin by applying initial stage factor analysis techniques, typically confirmatory factor analysis (CFA) in CB-SEM and confirmatory composite analysis (CCA) in PLS-SEM. They also assess complex models including higher-order constructs and provide mechanisms for exploring mediating and moderating effects within a single hypothesized model. Otherwise, CB-SEM and PLS-SEM have uniquely distinguishing characteristics. A fundamental difference between the two approaches lies in the representation of the latent constructs. Specifically, the measurement models represent constructs as *common factors* in CB-SEM and as *composites* in PLS-SEM (Hair & Sarstedt, 2020; Sarstedt et al., 2016). That is, in PLS-SEM constructs are represented by weighted composites of indicator variables with determinant construct scores thereby facilitating both in-sample and out-of-sample prediction (Hair, 2021; Shmueli et al., 2019).

5.1 | Global fit indices

While recent developments indicate metrics for the comparison of alternative competing PLS-SEM models are possible (see Liengaard et al., 2021), the full range of global goodness-of-fit measures utilized in CB-SEM is not available with

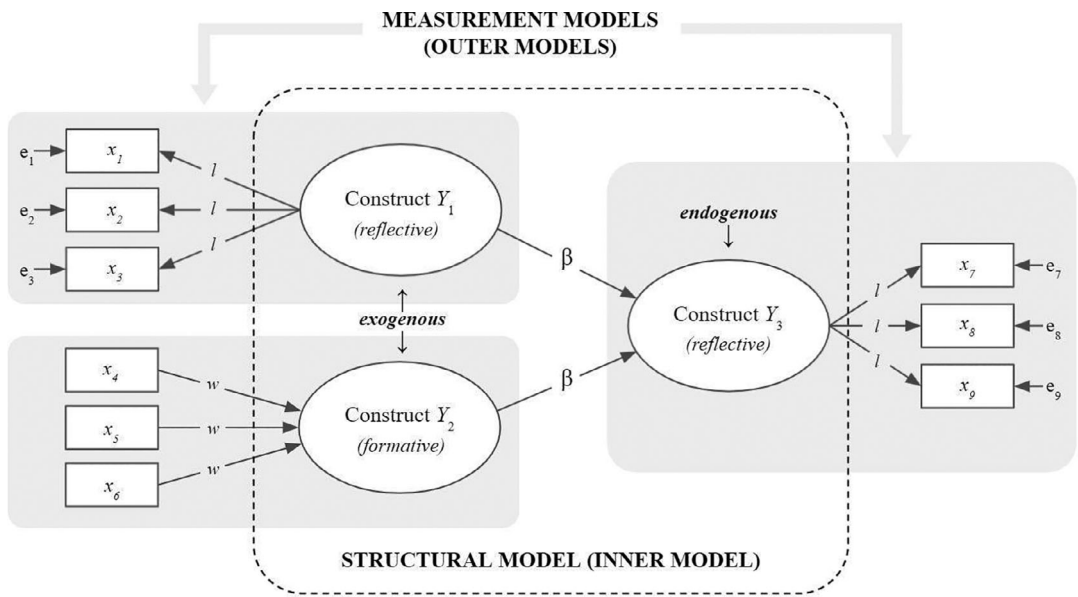


FIGURE 1 Simple PLS-SEM path model diagram. Reflective, reflective measurement model; Formative, formative measurement model; l , indicator outer loading; w , indicator outer weights; β , standardized path coefficient; e , error

PLS-SEM since model solutions are obtained based on a partial (rather than a full) information approach. Thus, in PLS-SEM applications, estimation is not based on addressing the gap between model-implied and data-implied covariance matrices or identifying a model that best fits the sample data. Rather, PLS-SEM model parameters such as indicator loadings and path coefficients are optimized based on the criterion of minimizing the unexplained variance (residuals) in the indicators and endogenous latent variables following an iterative process of estimating partial relationships within and between the constructs from successive approximations (Fornell & Bookstein, 1982; Hair et al., 2022).

5.2 | Treatment of measurement error

The treatment of measurement error in PLS-SEM also differs from CB-SEM. Recall that error terms capture the unexplained variance in constructs and indicators (i.e., items) when models are estimated (Schumacker & Lomax, 2016). Factor analysis procedures are commonly applied in research to reduce data, develop or refine instruments, and identify traits (Huck, 2012). Where CB-SEM applies CFA to test and potentially confirm measurement theory, PLS-SEM executes CCA to confirm composite-based measurement models (Hair, Howard, et al., 2020; Ringle et al., 2020). In estimating latent variables, PLS-SEM removes error variance by computing construct scores as linear combinations of the associated items (Hair, Black, et al., 2019). In other words, PLS-SEM calculates individual item weights, which are then used to compute construct scores. While error terms are not usually denoted in PLS-SEM structural model representations, the method explicitly accounts for measurement error inherent in the items (indicators) by including the assessment of item error in the measurement of the corresponding latent composite (Sarstedt et al., 2016).

5.3 | Formative measurement models

Assessment of formatively measured constructs is also guided by PLS-SEM's measurement philosophy (Sarstedt et al., 2016). Formatively measured constructs are increasingly common in business research as researchers attempt

to assess the direct impact of organizational practices on behavioral outcomes (Ringle et al., 2020). Reflective measurement assumes that a construct causes its observed indicators and is visualized with arrows that point from the construct to the indicators (Kline, 2016). In Figure 1, constructs Y_1 and Y_3 represent reflective measurement models. Alternatively, the indicators of formative measurement models are assumed to either cause or contribute to the underlying construct. Note that in the diagram, the arrows point from the indicators to the Y_2 construct. Thus, the construct is modeled as dependent on the observed indicators (Rigdon, 2016), and the path arrows are viewed much like regression paths since formative constructs are linear combinations of the indicators (Hair, Howard, et al., 2020). No error terms are associated with the individual indicators of the formative exogenous construct (x_4 , x_5 , and x_6), as shown in Figure 1. This is because formative causal indicators are assumed to be error-free. As formative indicators are conceptualized to capture the entirety of the construct, the decision to include formative measures in any model is driven by theory rather than by statistical analysis (Hair et al., 2022; Hair, Howard, et al., 2020; Hair, Sarstedt, et al., 2019). While PLS-SEM directly estimates composites representing formatively measured constructs, this kind of measurement model entails extremely restrictive assumptions in CB-SEM, which makes it technically very difficult to consider in applications (Diamantopoulos, 2011; Posey et al., 2015; Sarstedt et al., 2016).

6 | APPLICATION OF PLS-SEM: ASSESSMENT STAGES AND CRITERIA

To illustrate the assessment stages and guidelines for PLS-SEM, we examined a Commitment-Performance Model adapted from an applied study in the food services industry (See Figure 2; Hair, Page, et al., 2020). The dataset utilized contains partially simulated data to prevent identification of the company and demonstrate relevant concepts. The theoretical model in Figure 2 includes five exogenous constructs representing pay (PAY), supervision (SUPERVISION), training (TRAINING), team (TEAM), and self-development (SELFDEV), which are hypothesized to positively affect the endogenous construct organizational commitment (ORGCOMT). Organizational commitment, in turn, is hypothesized to positively affect the ultimate outcome variable of firm performance (PERFORMANCE).

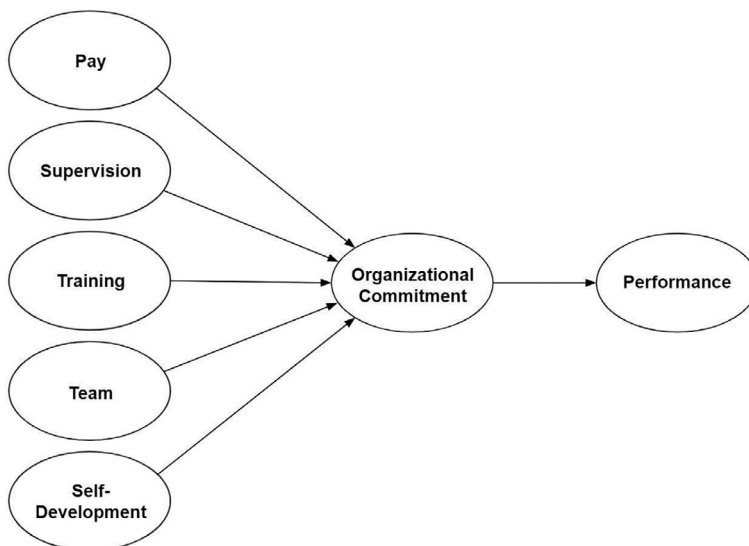


FIGURE 2 Theoretical model. Theoretical model for illustrative purposes. Adapted from J. F. Hair, M. Page, & N. Brunsveld (2020). *Essentials of Business Research Methods* (4th ed., p. 460), Taylor & Francis (<https://doi.org/10.4324/9780429203374>). Copyright 2020 by Taylor & Francis Group. Reprinted with permission

Four of the constructs were reflectively measured and included 3 items anchored on a 7-point scale (1 = *strongly disagree*, 7 = *strongly agree*). The training construct was measured formatively based on the number of times a particular type of training was completed (training_1 = *in-house training sessions*, training_2 = *online training sessions*, training_3 = *off-site training sessions*). The final endogenous outcome variable, performance, is represented by a single-item construct measured on a 7-point scale (1 = *very low performance*, 7 = *very high performance*) as reported by employee supervisors. Data ($n = 91$) were analyzed using SmartPLS 3 software (Ringle et al., 2015; Sarstedt & Cheah, 2019). Missing values were minimal and coded according to Hair et al. (2020). Following the inverse square root approach (Kock & Hadaya, 2018) and assuming the minimum path coefficient expected to be significant is between 0.21 and 0.30, one would need approximately 69 observations to render the corresponding effect significant at 5%. The 91 responses available can safeguard the analyses illustrated in this study. Table 1 provides a detailed list of assessment stages and recommended criteria for PLS-SEM.

6.1 | Evaluating measurement models

CB-SEM results are typically evaluated following a two-step process (Anderson & Gerbing, 1988). The first step assesses the measurement model by examining the validity and reliability of the constructs applying the CFA process. Once the measurement models are confirmed, the second step is evaluating the structural model. In PLS-SEM, the evaluation of measurement models is performed using CCA. In short, CCA is a systematic methodological process that involves sequentially examining relevant PLS-SEM metrics to evaluate the measurement and structural models. Following the steps in the CCA process enables researchers to fully evaluate the reliability and validity of both reflective and formative measurement models (Hair, Howard, et al., 2020). Steps in the CCA process (see Table 1) are similar to those in CFA but also include steps to evaluate formative measurement models.

6.1.1 | Initial reflective measurement model results

Figure 3 presents the theoretical model with initial estimation results. Indicator loadings for four of the five reflective multi-indicator constructs (PAY, SUPERVISION, TEAM, and ORGCOMT) were all above the recommended criteria (*loadings* ≥ 0.708 ; $p < 0.05$). For the SELFDEV construct, there were two potentially problematic items. Item selfdv_3 (*loading* = 0.623, $p = 0.001$) loaded slightly below the recommended threshold. Item selfdv_2 (*loading* = 0.139, $p = 0.291$) was substantially lower and was not statistically significant. In addition, if the two items were retained measures of internal consistency reliability, Cronbach's alpha (α), composite reliability (CR), and convergent validity, average variance extracted (AVE) would fall well below recommended guidelines ($\alpha = 0.431$; CR = 0.619; AVE = 0.418). As an initial effort to improve the self-development construct, the weakest item (selfdv_2) was removed. Prior to repeating the CCA for the updated reflective measurement models, we assessed the formative measurement model.

6.1.2 | Initial formative measurement model results

Following the CCA process, multicollinearity of the TRAINING construct was examined first. VIF values for all indicators were well below the threshold (≤ 2.20), indicating minimal multicollinearity among the indicators. The item training_1 outer weight size was quite small and did not meet significance thresholds ($\beta = 0.048$, $p = 0.436$). Formative indicators should not be removed from a measurement model based solely on statistical significance to ensure content validity is not sacrificed (Diamantopoulos & Winklhofer, 2001; Hair, Howard, et al., 2020). Therefore, we examined other criteria, including the contribution of the formative indicator as assessed by the size and significance of

TABLE 1 Assessment stages and criteria for PLS-SEM (CCA)

Assessment stage	Measure	Recommendation
Reflective measurement model		
Outer loading relevance testing	Indicator outer loadings	≥ 0.708 ; 95% CI
Construct internal consistency reliability	Composite reliability (CR); Cronbach's α	≥ 0.7 and ≤ 0.95
Convergent validity	Average variance extracted (AVE)	≥ 0.5
Discriminant validity	Heterotrait–monotrait ratio (HTMT)	< 0.85 to 0.90 ; 95% CI (one-sided)
Formative measurement model		
Convergent validity	Redundancy analysis	≥ 0.70 ; ideally ≥ 0.80
Assess for multicollinearity	Variance inflation factor (VIF)	< 3
Size and significance of indicators	Indicator outer weights	95% CI
Absolute contribution of indicators	Indicator outer loadings	≥ 0.50
Structural model		
Assess for multicollinearity	Variance inflation factor (VIF)	< 3
Size and significance of path coefficients	Standardized path coefficients (β)	closer to $ 1 $ = stronger; 95% CI
In-sample predictive ability	R^2 Endogenous variables	≥ 0.25 (weak); ≥ 0.50 (moderate); ≥ 0.75 (strong)
Effect size of predictive ability	f^2 Effect size	0.02 – 0.15 (small); 0.15 – 0.35 (medium); > 0.35 (large)
Out-of-sample predictive ability	Q^2_{predict} root mean squared error (RMSE)	prediction errors $<$ linear model naïve benchmark

the item loading. The training_1 item did not meet recommended guidelines and was removed from the training construct (Hair et al., 2022; Hair, Howard, et al., 2020).

6.1.3 | Respecified measurement model results

Following the deletion of two indicators (selfdv_2, reflective measurement model; training_1, formative measurement model), CCA was repeated to estimate the updated measurement models (Ringle et al., 2020). The results for the respecified reflective measurement model are presented in Table 2. The SELFDEV indicator selfdv_3 loaded slightly below the recommended threshold (0.627 , $p < 0.001$) but was statistically significant, and the construct exhibited validity as a two-item measure based on face-validity and theoretical relevance (Hair et al., 2022). The composite reliability for the SELFDEV construct is $.759$ which meets recommended guidelines. All remaining reflectively measured constructs exhibited internal consistency reliability ($\alpha = 0.819$ to 0.938 ; $CR = 0.882$ to 0.951 ; see Table 2). Convergent validity results for reflective constructs were above the critical value of 0.5 ($AVE \geq 0.620$). Discriminant validity, assessed with the heterotrait–monotrait criterion (HTMT; Henseler et al., 2015), indicated that all HTMT values were below the critical value of 0.9 (≤ 0.802). Moreover, the upper bound of the 95% percentile bootstrap confidence interval (one-sided) was below 0.90 for all HTMT results indicating that the HTMT values are significantly lower than the critical value (Franke & Sarstedt, 2019; Hair et al., 2022).

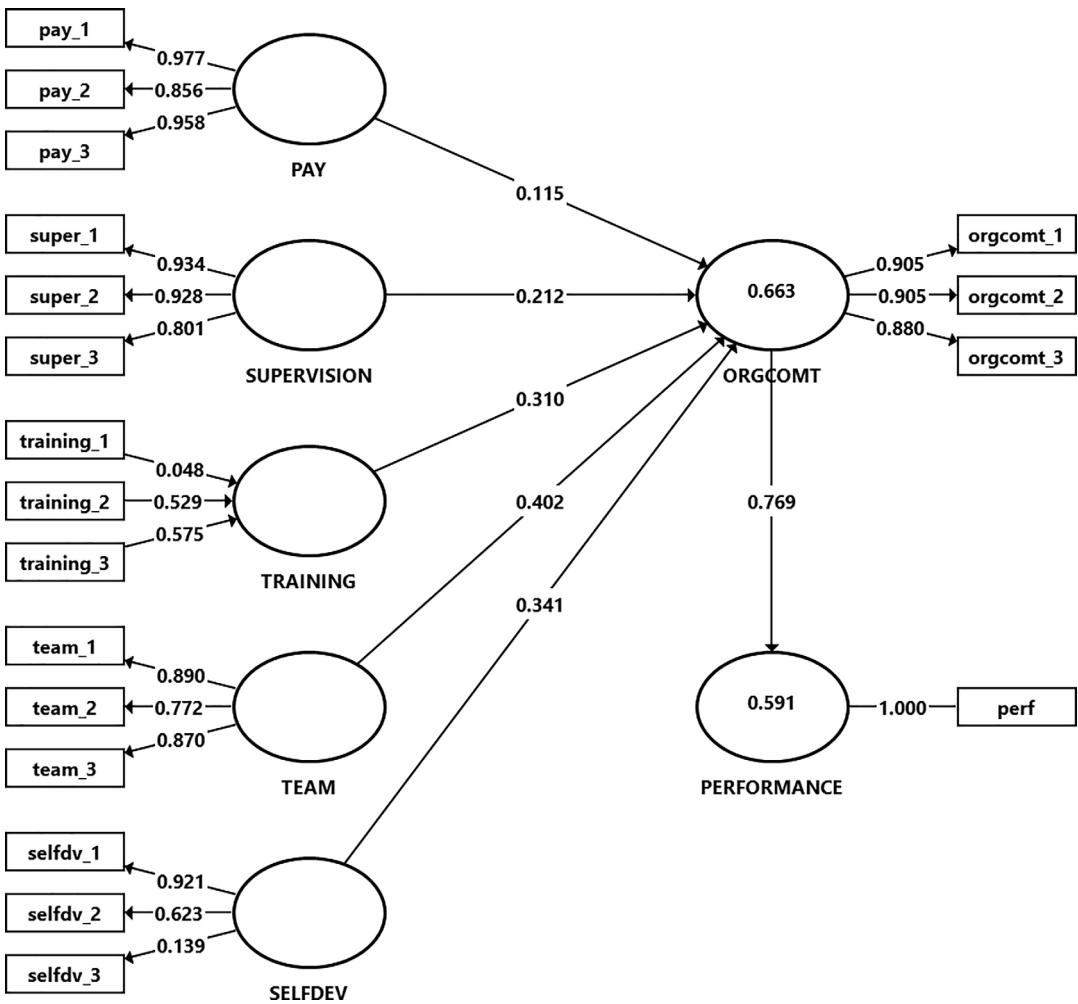


FIGURE 3 Initial model estimation 3

TABLE 2 CCA results for reflective measures

Variable	Cronbach's α	ρ_A	CR	AVE	Heterotrait–monotrait ratio (HTMT) ^a			
					2	3	4	5
1. ORGCOMT	0.878	0.884	0.925	0.804				
2. PAY	0.938	0.938	0.951	0.868	0.234			
3. SELFDEV	-		0.759	0.620	0.802	0.265		
4. SUPERVISION	0.866	0.865	0.919	0.792	0.503	0.102	0.514	
5. TEAM	0.819	0.877	0.882	0.715	0.521	0.122	0.226	0.220

Abbreviations: AVE, average variance extracted; CR, composite reliability; ORGCOMT, organizational commitment; SELFDEV, self-development.

^aThe upper bound of the 95% percentile bootstrap confidence interval (one-sided) is below 0.90 for all HTMT results.

The CCA formative model assessment steps were repeated for TRAINING by relating the formatively measured training construct with a global single-item construct. The results from the redundancy analysis ($\beta = 0.756$, $p < 0.001$) for the relationship between these two constructs were above the critical value; thus, convergent validity was confirmed. Likewise, collinearity was well below the recommended VIF value (1.378), and outer weights for training_2 ($\beta = 0.54$, $p = 0.004$) and training_3 ($\beta = 0.54$, $p < 0.002$) were meaningful and statistically significant. Significance testing used 95% percentile bootstrap confidence intervals (two-sided) and 10,000 bootstrap subsamples (Hair et al., 2022). In summary, both the reflective and formative measurement models met the recommended CCA guidelines (Hair, Howard, et al., 2020).

6.2 | Evaluating the structural model

When all measurement model guidelines have been met for CCA, the researcher next evaluates the relevance and predictive capability of the structural model. This involves a series of several steps to evaluate the model's explanatory and predictive elements. When evaluating a PLS-SEM model, researchers should (a) assess multicollinearity between endogenous constructs to ensure all inner VIF values are near or below 3; (b) examine the size and significance of the path coefficients in the structural model to verify all hypothesized relationships or predicted paths are meaningful and significant with p -values below 0.05; (c) assess in-sample predictive validity to ensure the coefficient of determination (R^2) measures are valid for all endogenous constructs using relevant research and contextual guidelines; and (d) assess out-of-sample predictive validity using $PLS_{predict}$.

Assessment metric results for the structural model are shown in Table 3. Collinearity was within the acceptable range ($VIF < 3$) for all constructs. Step two of CCA, assessing the size and significance of the path coefficients, revealed that all paths, with the exception of PAY, were significant ($p < 0.05$) with f^2 effect sizes ranging from medium to strong (see Table 3). The path coefficient of the PAY to ORGCOMT relationship was somewhat low ($\beta = 0.12$) but still within the acceptable range and was considered significant based on the small sample size ($n = 91$) and the specified directional hypothesis ($\pm$). As shown in Figure 4, the results from the illustrative model suggested that TEAM ($\beta = 0.40$, $f^2 = 0.47$) was the most important predictor of organizational commitment, followed by SELFDEV ($\beta = 0.34$, $f^2 = 0.29$) and TRAINING ($\beta = 0.31$, $f^2 = 0.27$). SUPERVISION ($\beta = 0.21$; $f^2 = 0.12$) and PAY ($\beta = 0.12$; $f^2 = 0.04$) had the smallest impact on organizational commitment. Overall, the illustrative model explained over 66% of the variance in ORGCOMT ($R^2 = 0.67$) and nearly 60% of the variance in PERFORMANCE ($R^2 = 0.59$), indicating moderate in-sample predictive capability (Hair, Risher, et al., 2019).

TABLE 3 Structural model results

Path	β	f^2	95% CI		p	VIF	R^2
			LL	UL			
Direct effects							
PAY → ORGCOMT	0.117	0.038	−0.078	0.214	0.080	1.083	0.665
SUPERVISION → ORGCOMT	0.214	0.115	0.091	0.330	0.002	1.193	
TRAINING → ORGCOMT	0.314	0.270	0.207	0.422	0.000	1.093	
TEAM → ORGCOMT	0.403	0.467	0.310	0.485	0.000	1.039	
SELFDEV → ORGCOMT	0.342	0.293	0.226	0.461	0.000	1.194	
ORGCOMT → PERFORMANCE	0.769	1.446	0.698	0.823	0.000		0.591

Abbreviations: CI, confidence interval (BCa); LL, lower limit; ORCOMT, organizational commitment; SELFDEV, self-development; UL, upper limit; VIF, variance inflation factor.

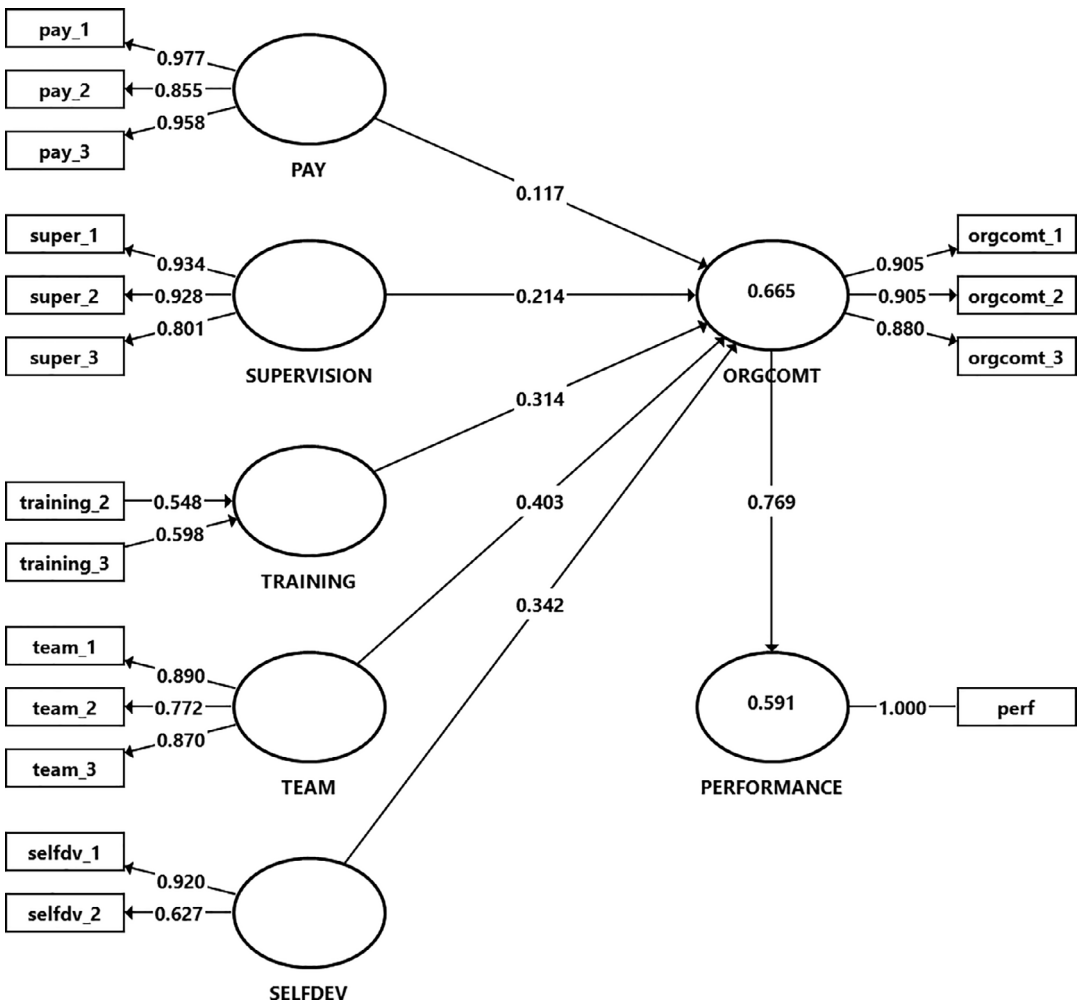


FIGURE 4 Final PLS-SEM results

6.3 | Evaluating out-of-sample prediction

Because the underlying PLS-SEM algorithm analyzes the data based on alternating partial calculations, global fit measures applied to structural model assessment in CB-SEM have not been fully developed for PLS-SEM (Hair et al., 2022). PLS-SEM, instead, enables researchers to assess the model's predictive capabilities by means of both in-sample and out-of-sample prediction, particularly with the $PLS_{predict}$ procedure (Shmueli et al., 2016). The $PLS_{predict}$ method divides the entire sample into k -folds, or subsets, then predicts the values of every subset with the remaining $k-1$ folds, thus treating each fold as a hold-out sample and the remaining folds as a training sample (Shmueli et al., 2019). These computations provide PLS-SEM prediction errors (i.e., for the indicators of endogenous constructs), which produce criteria such as the root mean squared error (RMSE) term. Alternative prediction approaches serve as benchmarks to assess these results. For instance, mean value prediction represents a naïve benchmark. If the $Q^2_{predict}$ value is positive, the corresponding prediction error is smaller than the naïve mean value prediction (Shmueli et al., 2019).

A more challenging benchmark is created with a direct linear model (LM), which regresses the indicators of the endogenous or predicted construct onto all indicators of the exogenous constructs in the model (Shmueli

TABLE 4 PLS_{predict} results

Variable	Q ² _{predict}	RMSE		ΔRMSE
		PLS-SEM model	LM model	
ORGCMT				
orgcomt_1	0.556	0.649	0.490	0.159
orgcomt_2	0.454	0.777	0.770	0.007
orgcomt_3	0.449	0.707	0.586	0.121
PERFORMANCE				
perf	0.491	1.274	1.285	−0.011

Abbreviations: LM, linear model (benchmark); ORGCMT, organizational commitment; RMSE, root mean squared error.

et al., 2016). When assessing the predictive power of the PLS-SEM model for a selected endogenous construct, the RMSE should be lower than the RMSE of the LM benchmark for the construct's indicators. If the RMSE for the PLS-SEM model is (a) lower than the RMSE for the LM for all items of this construct, then the model can be interpreted as having strong predictive capability; (b) is lower than that of the LM for most items, the model has moderate predictive capability; (c) outperforms the LM model for a few of the items, the result is low predictive capability; or (d) has higher RMSE values compared with the LM for all items, the model has poor predictive capability (Hair, Risher, et al., 2019).

Out-of-sample prediction was assessed using the PLS_{predict} procedure (Hair, 2021; Shmueli et al., 2016). As shown in Table 4, the model surpassed the naïve benchmark as the Q²_{predict} values were positive for the indicators of ORGCMT and PERFORMANCE. Moreover, for the indicator of the key target construct (i.e., PERFORMANCE), we found that the RMSE for the PLS-SEM model was smaller than that of the LM benchmark. Therefore, we can conclude that the model has strong external (out-of-sample) predictive capability (Hair et al., 2022; Shmueli et al., 2019). On the other hand, for ORGCMT's indicators, the RMSE of LM was lower than that of PLS-SEM; thus, the model had quite limited external predictive capability for this intermediate construct (see Table 4). While findings indicated moderate in-sample predictive capability, the results from the PLS_{predict} analysis suggested substantial out-of-sample predictive power for the PERFORMANCE construct (Hair et al., 2022; Shmueli et al., 2019).

7 | DISCUSSION

While the illustrative theoretical model is typical of organizational research in general, it contains several elements that analytical tools, such as multiple regression or CB-SEM, cannot (or cannot easily) accommodate. For example, the model contains multiple-item measures with weighted indicators and two dependent variables, which cannot be executed simultaneously with multiple linear regression (Sarstedt et al., 2020). The illustrative model also demonstrates the assessment of both formatively and reflectively measured constructs in the same model, which would require imposing additional model constraints in CB-SEM (see Posey et al., 2015). Lastly, the illustrative model demonstrates out-of-sample prediction assessment, which is possible with some other methods, but is more easily assessable with PLS software programs. Predictive assessments derived from in-sample prediction metrics (e.g., R²) can limit the conceptual and practical relevance of researchers' findings. As some scholars have noted, if we cannot predict successfully based on explanation, there is no basis for accepting the explanation, particularly since prediction is the primary criterion for evaluating theoretical falsifiability (Kaplan, 1964; Popper, 1962). Finding, for example, that the illustrative model demonstrated high out-of-sample prediction for the target firm performance outcome would be particularly valuable for providing managerial recommendations based on empirical PLS-SEM results.

This empirical example is not without limitations. First, results for the illustrative theoretical model (as adapted from Hair, Page, et al., 2020) were derived from a partially simulated dataset and, as such, are not intended for interpretation as conclusive applied findings. Second, while organizational commitment was modeled as an intervening variable, mediation was not assessed; therefore, conclusions cannot be drawn relative to causation. Third, a formative indicator was removed from the training construct. For formatively measured latent variables, items are not considered interchangeable; thus, the omission of any one item can substantially affect the validity of the measure (Hair, Risher, et al., 2019). This model was intended for illustrative purposes; however, in empirical applications, researchers can ensure with sufficient pre-tests that final analyses are not adversely impacted by the removal of formative indicators.

Lastly, several advanced analyses were not demonstrated in this introductory manuscript. Future contributions addressing additional features of PLS-SEM could further benefit the HRD research community. For example, it could be tested whether a theoretically established alternative model has higher predictive power and may be more suitable. The Bayesian information criterion (BIC) for predictive model comparison in PLS-SEM (Sharma et al., 2021) and the cross-validated predictive ability test (CVPAT; Liengaard et al., 2021) support comparative analytical assessment. Also, an importance-performance map analysis (IPMA; Ringle & Sarstedt, 2016) can offer relevant practical insights from PLS-SEM results. For a given target construct (e.g., organizational commitment), IPMA presents total effects and relative performance of predecessor constructs. In this way, scholar-practitioners could identify, for example, predictor constructs with low performance but high relative importance to prioritize and design interventions targeting organizational commitment.

In conclusion, the robust nature of PLS-SEM offers HRD researchers a unique set of advantages, including flexibility in data characteristics and model complexity, the ability to obtain suitable results using limited sample sizes relative to number of parameters, and comprehensive assessment of out-of-sample prediction (Hair, Risher, et al., 2019). Moreover, the application of the PLS-SEM methodology in the HRD field will enhance researchers' ability to achieve prediction-oriented goals in their research. Finally, integration of knowledge from related disciplines requires familiarity with research methodologies that have gained noticeable traction in tangent fields. To this end, cultivating awareness of PLS-SEM among HRD scholars and scholar-practitioners supports an increasingly essential multidisciplinary approach to organizational research (Torraco & Lundgren, 2020).

ACKNOWLEDGMENTS

The authors wish to thank Dr. Kim Nimon for encouraging this study and generously offering insight and feedback to improve earlier versions of the manuscript.

DATA AVAILABILITY STATEMENT

The data file that supports the illustrative model is available in the supplementary material of this article.

ORCID

Amanda E. Legate  <https://orcid.org/0000-0001-7763-7630>

Janice Lambert Chretien  <https://orcid.org/0000-0002-7979-4434>

REFERENCES

- Akdere, M., & Egan, T. (2020). Transformational leadership and human resource development: Linking employee learning, job satisfaction, and organizational performance. *Human Resource Development Quarterly*, 31(4), 393–421. <https://doi.org/10.1002/hrdq.21404>
- Anderson, J., & Gerbing, D. (1988). Structural equation modeling in practice: A review and recommended two-step approach. *Psychological Bulletin*, 103(3), 411–423. <https://doi.org/10.1037/0033-2909.103.3.411>
- Brown, S., & Leigh, T. (1996). A new look at psychological climate and its relationship to job involvement, effort, and performance. *Journal of Applied Psychology*, 81(4), 358–368. <https://doi.org/10.1037//0021-9010.81.4.358>

- Carter, J., & Youssef-Morgan, C. (2019). The positive psychology of mentoring: A longitudinal analysis of psychological capital development and performance in a formal mentoring program. *Human Resource Development Quarterly*, 30(3), 383–405. <https://doi.org/10.1002/hrdq.21348>
- Cepeda-Carrión, G., Henseler, J., Ringle, C., & Roldán, J. (2016). Prediction-oriented modeling in business research by means of PLS path modeling: Introduction to a JBR special section. *Journal of Business Research*, 69(10), 4545–4551. <https://doi.org/10.1016/j.jbusres.2016.03.048>
- Cepeda-Carrión G., Cegarra-Navarro J-G., Cillo V. (2019). Tips to use partial least squares structural equation modelling (PLS-SEM) in knowledge management. *Journal of Knowledge Management*, 23, (1), 67. –89. <http://doi.org/10.1108/jkm-05-2018-0322>
- Diamantopoulos, A. (2011). Incorporating formative measures into covariance-based structural equation models. *MIS Quarterly*, 35(2), 335–358. <https://doi.org/10.2307/23044046>
- Diamantopoulos, A., & Winklhofer, H. (2001). Index construction with formative indicators: An alternative to scale development. *Journal of Marketing Research*, 38(2), 269–277. <https://doi.org/10.1509/jmkr.38.2.269.18845>
- Dijkstra, T. (2010). Latent variables and indices: Herman Wold's basic design and partial least squares. In V. Esposito Vinzi, W. Chin, J. Henseler, & H. Wang (Eds.), *Handbook of partial least squares* (pp. 1–39). Springer. https://doi.org/10.1007/978-3-540-32827-8_2
- Fornell, C., & Bookstein, F. (1982). Two structural equation models: LISREL and PLS applied to consumer exit-voice theory. *Journal of Marketing Research*, 19(4), 440–452. <https://doi.org/10.2307/3151718>
- Fowler, F. (2014). *Survey research methods* (5th ed.). Sage.
- Franke, G., & Sarstedt, M. (2019). Heuristics versus statistics in discriminant validity testing: A comparison of four procedures. *Internet Research*, 29(3), 430–447. <https://doi.org/10.1108/IntR-12-2017-0515>
- Ghasemy, M., Teeroovengadam, V., Becker, J.-M., & Ringle, C. (2020). This fast car can move faster: A review of PLS-SEM application in higher education research. *Higher Education*, 80(6), 1121–1152. <https://doi.org/10.1007/s10734-020-00534-1>
- Goldberg, C., Rawski, S., & Perry, E. (2019). The direct and indirect effects of organizational tolerance for sexual harassment on the effectiveness of sexual harassment investigation training for HR managers. *Human Resource Development Quarterly*, 30(1), 81–100. <https://doi.org/10.1002/hrdq.21329>
- Goodhue, D., Lewis, W., & Thompson, R. (2012). Does PLS have advantages for small sample size or non-normal data? *MIS Quarterly*, 36(3), 981–1001. <https://doi.org/10.2307/41703490>
- Hair, J. (2021). Next generation prediction metrics for composite-based PLS-SEM. *Industrial Management & Data Systems*, 121(1), 5–11. <https://doi.org/10.1108/IMDS-08-2020-0505>
- Hair, J., Black, W., Babin, B., & Anderson, R. (2019). *Multivariate data analysis* (8th ed.). Cengage Learning.
- Hair, J., Howard, M., & Nitzl, C. (2020). Assessing measurement model quality in PLS-SEM using confirmatory composite analysis. *Journal of Business Research*, 109, 101–110. <https://doi.org/10.1016/j.jbusres.2019.11.069>
- Hair, J., Hult, G., Ringle, C., & Sarstedt, M. (2022). *A primer on partial least squares structural equation modeling (PLS-SEM)* (3rd ed.). Sage.
- Hair, J., Matthews, L., Matthews, R., & Sarstedt, M. (2017). PLS-SEM or CB-SEM: Updated guidelines on which method to use. *International Journal of Multivariate Data Analysis*, 1(2), 107–123. <https://doi.org/10.1504/ijmda.2017.087624>
- Hair, J., Page, M., & Brunsveld, N. (2020). *Essentials of business research methods* (4th ed.). Routledge.
- Hair, J., Ringle, C., & Sarstedt, M. (2011). PLS-SEM: Indeed a silver bullet. *Journal of Marketing Theory and Practice*, 19(2), 139–151. <https://doi.org/10.1080/10696679.2011.11046435>
- Hair, J., Risher, J., Sarstedt, M., & Ringle, C. (2019). When to use and how to report the results of PLS-SEM. *European Business Review*, 31(1), 2–24. <https://doi.org/10.1108/EBR-11-2018-0203>
- Hair, J., & Sarstedt, M. (2020). Factors versus composites: Guidelines for choosing the right structural equation modeling method. *Project Management Journal*, 50(6), 619–624. <https://doi.org/10.1177/8756972819882132>
- Hair, J., & Sarstedt, M. (2021). Explanation plus prediction – The logical focus of project management research. *Project Management Journal*. Advance online publication, 52(4), 319–322. <https://doi.org/10.1177/8756972821999945>
- Hair, J., Sarstedt, M., Pieper, T., & Ringle, C. (2012). The use of partial least squares structural equation modeling in strategic management research: A review of past practices and recommendations for future applications. *Long Range Planning*, 45(5–6), 320–340. <https://doi.org/10.1016/j.lrp.2012.09.008>
- Hair, J., Sarstedt, M., & Ringle, C. (2019). Rethinking some of the rethinking of partial least squares. *European Journal of Marketing*, 53(4), 566–584. <https://doi.org/10.1108/EJM-10-2018-0665>
- Hayes, A., & Rockwood, N. (2020). Conditional process analysis: Concepts, computation, and advances in the modeling of the contingencies of mechanisms. *The American Behavioral Scientist*, 64(1), 19–54. <https://doi.org/10.1177/0002764219859633>
- Hendry, D., Morgan, M., & Wold, H. (1994). The ET interview: Professor H. O. A. Wold: 1908–1992. *Econometric Theory*, 10(2), 419–433. <https://www.jstor.org/stable/3532876>

- Henseler, J., Ringle, C., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, 43(1), 115–135.
- Huck, S. W. (2012). *Reading statistics and research* (6th ed.). Pearson.
- Hutchins, H., Penney, L., & Sublett, L. (2018). What imposters risk at work: Exploring imposter phenomenon, stress coping, and job outcomes. *Human Resource Development Quarterly*, 29(1), 31–48. <https://doi.org/10.1002/hrdq.21304>
- Jeong, S. (2021). A cross-level analysis of organizational knowledge creation: How do transformational leaders interact with their subordinates' expertise and interpersonal relationships? *Human Resource Development Quarterly*, 32(2), 111–130. <https://doi.org/10.1002/hrdq.21416>
- Jöreskog, K., & Wold, H. (1982). The ML and PLS techniques for modeling with latent variables: Historical and comparative aspects. In H. Wold & K. G. Jöreskog (Eds.), *Systems under indirect observation, Part I* (pp. 263–270). North-Holland.
- Kaplan, A. (1964). *The conduct of inquiry: Methodology for behavioural science*. Chandler Publishing.
- Khan, G., Sarstedt, M., Shiau, W., Hair, J., Ringle, C., & Fritze, M. (2019). Methodological research on partial least squares structural equation modeling (PLS-SEM). *Internet Research*, 29(3), 407–429. <https://doi.org/10.1108/IntR-12-2017-0509>
- Kline, R. B. (2016). *Principles and practice of structural equation modeling* (4th ed.). Guilford Press.
- Kock, N., & Hadaya, P. (2018). Minimum sample size estimation in PLS-SEM: The inverse square root and gamma-exponential methods. *Information Systems Journal*, 28(1), 227–261. <https://doi.org/10.1111/isj.12131>
- Lee, L., Petter, S., Fayard, D., & Robinson, S. (2011). On the use of partial least squares path modeling in accounting research. *International Journal of Accounting Information Systems*, 12(4), 305–328. <https://doi.org/10.1016/j.accinf.2011.05.002>
- Lienggaard, B., Sharma, P., Hult, G., Jensen, M., Sarstedt, M., Hair, J., & Ringle, C. (2021). Prediction: Coveted, yet forsaken? Introducing a cross-validated predictive ability test in partial least squares path modeling. *Decision Sciences*, 52(2), 362–392. <https://doi.org/10.1111/deci.12445>
- Lohmöller, J.-B. (1989). *Latent variable path modeling with partial least squares*. Springer.
- Manley, S., Hair, J., Williams, R., & McDowell, W. (2020). Essential new PLS-SEM analysis methods for your entrepreneurship analytical toolbox. *International Entrepreneurship and Management Journal*. <https://doi.org/10.1007/s11365-020-00687-6>
- Marcoulides, G., Chin, W., & Saunders, C. (2012). When imprecise statistical statements become problematic: A response to Goodhue, Lewis, and Thompson. *MIS Quarterly*, 36(3), 717–728. <https://doi.org/10.2307/41703477>
- McIntosh, C., Edwards, J., & Antonakis, J. (2014). Reflections on partial least squares path modeling. *Organizational Research Methods*, 17(2), 210–251. <https://doi.org/10.1177/1094428114529165>
- Nathans, L., Oswald, F., & Nimon, K. (2012). Interpreting multiple linear regression: A guidebook of variable importance. *Practical Assessment, Research & Evaluation*, 17(9), 1–19. <https://doi.org/10.7275/5fex-b874>
- Nimon, K. (2012). Statistical assumptions of substantive analyses across the general linear model: A mini-review. *Frontiers in Psychology*, 3, 322. <https://doi.org/10.3389/fpsyg.2012.00322>
- Nitzl, C. (2016). The use of partial least squares structural equation modelling (PLS-SEM) in management accounting research: Directions for future theory development. *Journal of Accounting Literature*, 37, 19–35. <https://doi.org/10.1016/j.acclit.2016.09.003>
- Osam, K., Shuck, B., & Immekus, J. (2020). Happiness and healthiness: A replication study. *Human Resource Development Quarterly*, 31(1), 75–89. <https://doi.org/10.1002/hrdq.21373>
- Podsakoff, P., MacKenzie, S., Lee, J., & Podsakoff, N. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88(5), 879–903. <https://doi.org/10.1037/0021-9010.88.5.879>
- Popper, K. R. (1962). *Conjectures and refutations: The growth of scientific knowledge*. Basic Books.
- Posey, C., Roberts, T., Lowry, P., & Bennett, R. (2015). Multiple indicators and multiple cause (MIMIC) models as a mixed-modeling technique: A tutorial and annotated example. *Communications of the Association for Information Systems*, 36(1), 179–204. <https://doi.org/10.17705/1cais.03611>
- Richter, N., Sinkovics, R., Ringle, C., & Schlägel, C. (2016). A critical look at the use of SEM in international business research. *International Marketing Review*, 33(3), 376–404. <https://doi.org/10.1108/IMR-04-2014-0148>
- Rigdon, E. (2016). Choosing PLS path modeling as analytical method in European management research: A realist perspective. *European Management Journal*, 34(6), 598–605. <https://doi.org/10.1016/j.emj.2016.05.006>
- Rigdon, E., Sarstedt, M., & Ringle, C. (2017). On comparing results from CB-SEM and PLS-SEM: Five perspectives and five recommendations. *Marketing ZFP*, 39(3), 4–16. <https://doi.org/10.15358/0344-1369-2017-3-4>
- Ringle, C., Sarstedt, M., Mitchell, R., & Gudergan, S. (2020). Partial least squares structural equation modeling in HRM research. *International Journal of Human Resource Management*, 31(12), 1617–1643. <https://doi.org/10.1080/09585192.2017.1416655>

- Ringle, C., Sarstedt, M., & Straub, D. (2012). A critical look at the use of PLS-SEM in MIS quarterly. *MIS Quarterly*, 36(1), iii–xiv. <https://doi.org/10.2307/41410402>
- Ringle, C., Wende, S. & Becker J.-M. (2015). SmartPLS 3 [Computer software]. Bönningstedt: SmartPLS. <https://www.smartpls.com/>
- Ringle C. M., Sarstedt M. (2016). Gain more insight from your PLS-SEM results: The importance-performance map analysis. *Industrial Management & Data Systems*, 116, (9), 1865. –1886. <http://doi.org/10.1108/imds-10-2015-0449>
- Rönkkö, M., & Evermann, J. (2013). A critical examination of common beliefs about partial least squares path modeling. *Organizational Research Methods*, 16(3), 425–448. <https://doi.org/10.1177/1094428112474693>
- Rönkkö, M., McIntosh, C., Antonakis, J., & Edwards, J. (2016). Partial least squares path modeling: Time for some serious second thoughts. *Journal of Operations Management*, 47–48(1), 9–27. <https://doi.org/10.1016/j.jom.2016.05.002>
- Sarstedt, M., & Cheah, J. (2019). Partial least squares structural equation modeling using SmartPLS: A software review. *Journal of Marketing Analytics*, 7(3), 196–202. <https://doi.org/10.1057/s41270-019-00058-3>
- Sarstedt, M., Hair, J., Cheah, J., Becker, J., & Ringle, C. (2019). How to specify, estimate, and validate higher-order constructs in PLS-SEM. *Australasian Marketing Journal*, 27(3), 197–211. <https://doi.org/10.1016/j.ausmj.2019.05.003>
- Sarstedt, M., Hair, J., Nitzl, C., Ringle, C., & Howard, M. (2020). Beyond a tandem analysis of SEM and PROCESS: Use PLS-SEM for mediation analyses! *International Journal of Market Research*, 62(3), 288–299. <https://doi.org/10.1177/1470785320915686>
- Sarstedt, M., Hair, J., Ringle, C., Thiele, K., & Gudergan, S. (2016). Estimation issues with PLS and CBSEM: Where the bias lies! *Journal of Business Research*, 69(10), 3998–4010. <https://doi.org/10.1016/j.jbusres.2016.06.007>
- Sarstedt, M., Ringle, C., & Hair, J. (2014). PLS-SEM: Looking back and moving forward. *Long Range Planning*, 47(3), 132–137. <https://doi.org/10.1016/j.lrp.2014.02.008>
- Sarstedt, M., Ringle, C., Smith, D., Reams, R., & Hair, J. (2014). Partial least squares structural equation modeling (PLS-SEM): A useful tool for family business researchers. *Journal of Family Business Strategy*, 5(1), 105–115. <https://doi.org/10.1016/j.jfbs.2014.01.002>
- Schumacker, R., & Lomax, R. (2016). *A beginner's guide to structural equation modeling* (4th ed.). Taylor & Francis.
- Sharma, P., Shmueli, G., Sarstedt, M., Danks, N., & Ray, S. (2021). Prediction-oriented model selection in partial least squares path modeling. *Decision Sciences*, 52(3), 567–607. <https://doi.org/10.1111/dec.12329>
- Shiau, W., Sarstedt, M., & Hair, J. (2019). Internet research using partial least squares structural equation modeling (PLS-SEM). *Internet Research*, 29(3), 398–406. <https://doi.org/10.1108/IntR-10-2018-0447>
- Shmueli, G. (2010). To explain or to predict? *Statistical Science*, 25(3), 289–310. <https://doi.org/10.1214/10-sts330>
- Shmueli, G., Ray, S., Velasquez Estrada, J., & Shatla, S. (2016). The elephant in the room: Evaluating the predictive performance of PLS models. *Journal of Business Research*, 69(10), 4552–4564. <https://doi.org/10.1016/j.jbusres.2016.03.049>
- Shmueli, G., Sarstedt, M., Hair, J., Cheah, J., Ting, H., Vaithilingam, S., & Ringle, C. (2019). Predictive model assessment in PLS-SEM: Guidelines for using PLSpredict. *European Journal of Marketing*, 53(11), 2322–2347. <https://doi.org/10.1108/EJM-02-2019-0189>
- Streukens, S., & Leroi-Werelds, S. (2016). Bootstrapping and PLS-SEM: A step-by-step guide to get more out of your bootstrap results. *European Management Journal*, 34(6), 618–632. <https://doi.org/10.1016/j.emj.2016.06.003>
- Takeuchi, T., Takeuchi, N., & Jung, Y. (2020). Toward a process model of newcomer socialization: Integrating pre- and post-entry factors for newcomer adjustment. *Human Resource Development Quarterly*, 32(3), 391–418. <https://doi.org/10.1002/hrdq.21420>
- Torraco, R., & Lundgren, H. (2020). What HRD is doing—What HRD should be doing: The case for transforming HRD. *Human Resource Development Review*, 19(1), 39–65. <https://doi.org/10.1177/1534484319877058>
- Willaby, H., Costa, D., Burns, B., MacCann, C., & Roberts, R. (2015). Testing complex models with small sample sizes: A historical overview and empirical demonstration of what partial least squares (PLS) can offer differential psychology. *Personality and Individual Differences*, 84(1), 73–78. <https://doi.org/10.1016/j.paid.2014.09.008>
- Wold, H. (1982). Soft modeling: The basic design and some extensions. In K. G. Jöreskog & H. Wold (Eds.), *Systems under indirect observations: Part II* (pp. 1–54). North-Holland.
- Zhao, X., Lynch, J., & Chen, Q. (2010). Reconsidering Baron and Kenny: Myths and truths about mediation analysis. *Journal of Consumer Research*, 37(2), 197–206. <https://doi.org/10.1086/651257>
- Zigarmi, D., Roberts, T., & Randolph, A. (2015). Employee's perceived use of leader power and implications for affect and work intentions. *Human Resource Development Quarterly*, 26(4), 359–384. <https://doi.org/10.1002/hrdq.21216>

AUTHOR BIOGRAPHIES

Amanda Legate is a doctoral student in the Human Resource Development (HRD) PhD program at the University of Texas at Tyler. Professionally, she serves as the HRD officer for a technology and cybersecurity firm headquartered in central Arkansas. Her primary research interests include applied psychometrics and statistical modelling relevant to organizational and workforce agility.

Dr. Joe F. Hair, Jr. holds the Cleverdon Chair of Business, Mitchell College of Business, University of South Alabama, U.S.A. Joe is ranked in the top 1% globally of all Business and Economics professors by Clarivate Analytics based on his citations and scholarly accomplishments, which for his career exceed 289,000. He has authored over 80 editions of his books, which include *Multivariate Data Analysis*, Cengage Learning, U.K., 8th edition, 2019 (cited 166,000+ times and one of the top three all time social sciences research methods textbooks); *A Primer on Partial Least Squares Structural Equation Modeling*, Sage, 3rd edition, 2022; a SEMnR version of the Primer, Springer, 2022; *Essentials of Business Research Methods*, Routledge, 4th edition 2020; and *Essentials of Marketing Analytics*, McGraw-Hill, 2021. He also has published numerous articles in peer reviewed scholarly journals, and has a new book on Sales Analytics, forthcoming in 2022 (Chicago Business Press).

Dr. Janice Lambert Chretien is an adjunct instructor of statistics at The University of Texas at Tyler Soules College of Business. Her research embraces instructional design, learning transfer, memory, and neurodiversity.

Dr. Jeffrey Risher is an Assistant Professor of Quantitative Methods at Southeastern Oklahoma State University. His primary research areas are Supply Chain Logistics and Data Analytics. His additional research interests include Customer Relationship Management and Marketing Strategy, paying specific attention to the role of analytics in demand and supply integration.

How to cite this article: Legate, A. E., Hair, J. F. Jr, Chretien, J. L., & Risher, J. J. (2023). PLS-SEM: Prediction-oriented solutions for HRD researchers. *Human Resource Development Quarterly*, 34(1), 91–109. <https://doi.org/10.1002/hrdq.21466>